RAY KWAN

SOFTWARE ENGINEER

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SUMMARY

Software Engineer with 7+ years of experience building precise manufacturing systems and full-stack software. Proven record of leading teams to deliver high-performance CNC and laser vision solutions adopted by numerous NHL teams

WORK EXPERIENCE

SOFTWARE ENGINEER
Jan 2026 - present

BAUER HOCKEY

- Evaluated and selected scanning and CNC control software for **Bauer's next-generation** skate tuning machine.
- Integrated Aetrex APIs into Skatescribe's skate tuning app, automating customer data transfer and eliminating manual entry for operators
- **Horizontally scaled** existing cloud infrastructure to support 700+ machines globally

LEAD ENGINEER (promoted from FULL STACK DEV)
May 2021 - present

SKATESCRIBE

- Introduced **±1 µm** precision to skate sharpening by leading 7 engineers to design and build a laser-integrated CNC system, now adopted by **6+ NHL teams**.
- Architected to **microservices** refactor to decouple core business logic, decreasing cycle time by **25%** and improving data accuracy by **70%**.
- Deployed scalable cloud infrastructure with **Docker, NGINX,** and **Linode**.

FULL STACK DEVELOPER
Sept 2020 - May 2021

COGNITIVE CENTIVIZER

- Developed **React** and **Node.js** games to measure reaction time and memory, integrating real-time data collection and analytics for 100+ participants.

SOFTWARE ENGINEER
Jan 2021 - April 2021

OPENRACE – UofT Hatchery

- Built a virtual racing app in **Swift** and **NodeJS** that was funded by the **UofT Hatchery** accelerator program

FULL STACK DEVELOPER
May 2019 - June 2020

AUTOCASE

- Developed a cost-benefit analysis platform (**AngularJS** and **Python**) transforming economic research models into interactive visualizations of building design impacts.

SKILLS

Languages & Frameworks
- Python
- Flask
- Javascript
- ReactJS/ NextJS
- AngularJS
- NodeJS
- Java
- PSQL

Libraries
- Numpy/ Scipy
- Pandas
- MUI

DevOps
- Docker
- Linode
- AWS

WEBSITES

- Website
- LinkedIn

Projects

- Music Player
- Wordle Copycat
- Graphing Library
- ATM

EDUCATION

B.Sc. Computer Science
University of Toronto
2016 - 2021
Dean's Honor (**CGPA: 3.72/4**)

Digital Growth Marketing
Talent Accelerator